

The lesion is dosage insufficiency, not a druggable target

Compound heterozygous BUB1B

p.Leu737Ter — deletes the pseudokinase domain

p.Asn1002Lys — buried inside it, 20% accessible

You cannot inhibit your way out of an insufficiency.

No drug target on BubR1 — three independent lines

Literature 13/16 papers: vertebrate BubR1 is a pseudokinase

Structure missense is 34.7 Å from the nucleotide site

Open Targets 28 tractability buckets, no pocket of any quality

This is why no docking was performed.

A mouse carries this child's allelic architecture

BubR1^{H/L1002P} = reduced protein + C-lobe missense

Full MVA phenotype: cancer, short lifespan, sarcopenia

Progeroid severity tracked with SASP complexity

→ Dasatinib: FDA approved, pediatric from age 1,
one indication in combination with chemotherapy

Readthrough: right codon, wrong product

Stop created = TGA-A — the most permissive stop class

UGA readthrough inserts Trp, Cys or Arg — not Leu

Position 737 across 247 vertebrate orthologs:

Leu 236	Trp 0	Cys 0	Arg 0
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90% buried. We demoted our own leading candidate.

What we argue against

Aneuploid tumours depend on residual checkpoint —
so checkpoint inhibitors are the standard play.

Not here: this child's NORMAL cells are checkpoint-
compromised too. The therapeutic window may be absent.

Spindle poisons: 30 papers 'more sensitive', 18 'more
resistant'. Nothing justifies altering his chemotherapy.

Generalises — and it is a hypothesis

ptc_triage.py — any nonsense variant in, triage out:

stop class · +4 context · NMD competence ·

ortholog tolerance of the readthrough products

Applies to thousands of recessive diseases.

Next step is a dish, not a prescription.